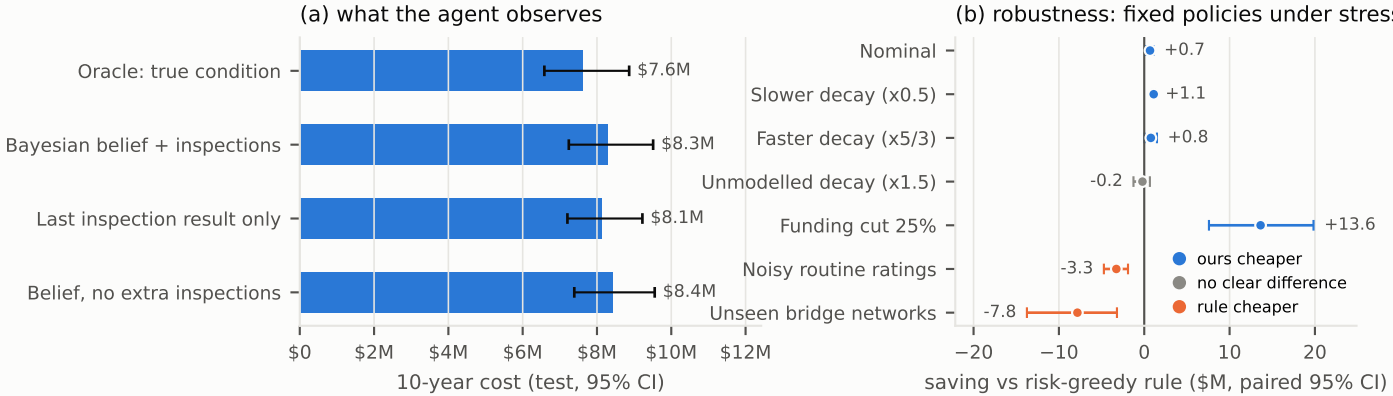
(a) what the agent observes
Oracle: true condition
Bayesian belief + inspections
Last inspection result only
Belief, no extra inspections
$7.6M
$8.3M
$8.1M
$8.4M
$0
$2M
$4M
$6M
$8M
$10M
$12M
10-year cost (test, 95% CI)
(b) robustness: fixed policies under stress
Nominal
Slower decay (x0.5)
Faster decay (x5/3)
Unmodelled decay (x1.5)
Funding cut 25%
Noisy routine ratings
Unseen bridge networks
+0.7
+1.1
+0.8
-0.2
+13.6
-3.3
-7.8
ours cheaper
no clear difference
rule cheaper
−20
−10
0
10
20
saving vs risk-greedy rule ($M, paired 95% CI)